Find

This is a quick reference list of cheatsheet for linux find command, contains common options and examples.

Getting Started

Usage

```
$ find [path...] [options] [expression]
```

Wildcard

```
$ find . -name "*.txt"
$ find . -name "2020*.csv"
$ find . -name "json_*"
```

- Regex reference (quickref.me)
- Find cheatsheet (gist.github.com)

Option Examples

Option	Example	Description
`-type`	find . -type d	Find only directories
`-name`	find . -type f -name "*.txt"	Find file by name
`-iname`	find . -type f -iname "hello"	Find file by name (case-insensitive)
`-size`	find . -size +1G	Find files larger than 1G

Option	Example	Description
`-user`	find . -type d -user jack	Find jack's file
`-regex`	find /var -regex '.*/tmp/.*[0-9]*.file'	Using Regex with find. See regex
`-maxdepth`	find . -maxdepth 1 -name "a.txt"	In the current directory and subdirectories
`-mindepth`	find / -mindepth 3 -maxdepth 5 -name pass	Between sub-directory level 2 and 4

Type

`-type d`	Directory
`-type f`	File
`-type l`	Symbolic link
`-type b`	Buffered block
`-type c`	Unbuffered character
`-type p`	Named pipe
`-type s`	Socket

Size

`-size b`	512-byte blocks (default)
`-size c`	Bytes
`-size k`	Kilobytes
`-size M`	Megabytes
`-size G`	Gigabytes
`-size T`	Terabytes (only BSD)
`-size P`	Petabytes (only BSD)

Size +/-

Find all bigger than 10MB files

```
$ find / -size +10M
```

Find all smaller than 10MB files

```
$ find / -size -10M
```

Find all files that are exactly 10M

```
$ find / -size 10M
```

Find Size between 100MB and 1GB

```
$ find / -size +100M -size -1G
```

The + and - prefixes signify greater than and less than, as usual.

Names

Find files using name in current directory

```
$ find . -name tecmint.txt
```

Find files under home directory

```
$ find /home -name tecmint.txt
```

Find files using name and ignoring case

```
$ find /home -iname tecmint.txt
```

Find directories using name

```
$ find / -type d -name tecmint
```

Find php files using name

```
$ find . -type f -name tecmint.php
```

Find all php files in directory

```
$ find . -type f -name "*.php"
```

Permissions

Find the files whose permissions are 777.

```
$ find . -type f -perm 0777 -print
```

Find the files without permission 777.

```
$ find / -type f ! -perm 777
```

Find SUID set files.

```
$ find / -perm /u=s
```

Find SGID set files.

```
$ find / -perm /g=s
```

Find Read Only files.

```
$ find / -perm /u=r
```

Find Executable files.

```
$ find / -perm /a=x
```

Owners and Groups

Find single file based on user

```
$ find / -user root -name tecmint.txt
```

Find all files based on user

```
$ find /home -user tecmint
```

Find all files based on group

```
$ find /home -group developer
```

Find particular files of user

```
$ find /home -user tecmint -iname "*.txt"
```

Multiple filenames

```
$ find . -type f \( -name "*.sh" -o -name "*.txt" \)
```

Find files with `.sh` and `.txt` extensions

Multiple dirs

```
$ find /opt /usr /var -name foo.scala -type f
```

Find files with multiple dirs

Empty

```
$ find . -type d -empty
```

Delete all empty files in a directory

```
$ find . -type f -empty -delete
```

Find Date and Time

	Means
`atime`	access time (last time file opened)
`mtime`	modified time (last time file contents was modified)
`ctime`	changed time (last time file inode was changed)
	Example
`-mtime +0`	Modified greater than 24 hours ago
`-mtime 0`	Modified between now and 1 day ago
`-mtime -1`	Modified less than 1 day ago (same as `-mtime 0`)

`-mtime 1`	Modified between 24 and 48 hours ago
`-mtime +1`	Modified more than 48 hours ago
`-mtime +1w`	Last modified more than 1 week ago
`-atime 0`	Last accessed between now and 24 hours ago
`-atime +0`	Accessed more than 24 hours ago
`-atime 1`	Accessed between 24 and 48 hours ago
`-atime +1`	Accessed more than 48 hours ago
`-atime -1`	Accessed less than 24 hours ago (same as `-atime 0`)
`-ctime -6h30m`	File status changed within the last 6 hours and 30 minutes

Examples

Find last 50 days modified files

```
$ find / -mtime 50
```

find last 50 days accessed files

```
$ find / -atime 50
```

find last 50-100 days modified files

```
$ find / -mtime +50 –mtime -100
```

find changed files in last 1 hour

```
$ find / -cmin -60
```

find modified files in last 1 hour

```
$ find / -mmin -60
```

find accessed files in last 1 hour

```
$ find / -amin -60
```

Find and

Find and delete

Find and remove multiple files

```
$ find . -type f -name "*.mp3" -exec rm -f {} \;
```

Find and remove single file

```
$ find . -type f -name "tecmint.txt" -exec rm -f {} \;
```

Find and delete 100mb files

```
$ find / -type f -size +100m -exec rm -f {} \;
```

Find specific files and delete

```
$ find / -type f -name *.mp3 -size +10m -exec rm {} \;
```

Find and replace

Find all files and modify the content `const` to `let`

```
$ find ./ -type f -exec sed -i 's/const/let/g' {} \;
```

Find readable and writable files and modify the content `old` to `new`

```
$ find ./ -type f -readable -writable -exec sed -i "s/old/new/g" {} \;
```

See also: sed cheatsheet

Find and rename

Find and suffix (added `.bak`)

```
$ find . -type f -name 'file*' -exec mv {} {}.bak\;
```

Find and rename extension (`.html` => `.gohtml`)

```
$ find ./ -depth -name "*.html" -exec sh -c 'mv "$1" "${1%.html}.gohtml"' _ {} \;
```

Find and move

```
$ find . -name '*.mp3' -exec mv {} /tmp/music \;
```

Find and move it to a specific directory (`/tmp/music`)

Find and copy

```
$ find . -name '*2020*.xml' -exec cp -r "{}" /tmp/backup \;
```

Find matching files and copy to a specific directory (`/tmp/backup`)

Find and concatenate

Merge all csv files in the download directory into `merged.csv`

```
$ find download -type f -iname '*.csv' | xargs cat > merged.csv
```

Merge all sorted csv files in the download directory into `merged.csv`

```
$ find download -type f -iname '*.csv' | sort | xargs cat > merged.csv
```

Find and sort

Find and sort in ascending

```
$ find . -type f | sort
```

find and sort descending

```
$ find . -type f | sort -r
```

Find and chmod

Find files and set permissions to 644.

```
$ find / -type f -perm 0777 -print -exec chmod 644 {} \;
```

Find directories and set permissions to 755.

```
$ find / -type d -perm 777 -print -exec chmod 755 {} \;
```

Find and compress

Find all `.java` files and compress it into `java.tar`

```
$ find . -type f -name "*.java" | xargs tar cvf java.tar
```

Find all `.csv` files and compress it into `quickref.zip`

```
$ find . -type f -name "*.csv" | xargs zip quickref.zip
```

Related Cheatsheet

Grep Cheatsheet
Quick Reference

Recent Cheatsheet

Remote Work Revolution Cheatsheet
Quick Reference

Homebrew Cheatsheet
Quick Reference

PyTorch Cheatsheet
Quick Reference

Taskset Cheatsheet
Quick Reference